

TABLE IV
Streamlining of Binary Table

IDS equivalent	Existing binary notation	New name	Pronunciation	New figure
0	0	Just O	As in owe	⓪
1	1	Just I	As in eye	I

TABLE V
Example for Streamlining of Binary Table

Old method, figures and words	New figures and how to read them
10101 → one zero one zero one	I Φ I Φ I → I-o-I-o-I
1100.101 → one one zero zero point one zero one	I I Φ Φ, I Φ I → I-I-o-o-point-I-o-I

TABLE VI
Octal Table of Single-digit Number

IDS equivalent	Octal No.	New figure	Old name	Read new name as
0	0	0	Zero	No change in reading; read as zero only
1	1	1	One	Read as one
2	2	2	Two	Read as two
3	3	3	Three	Read as three
4	4	4	Four	Read as four
5	5	5	Five	Read as five
6	6	6	Six	Read as six
7	7	7	Seven	Read as seven
8	10	10	?	There is change from now on. Read as one zero; with a dash in between as 1-0. The number ten does not exist in this system, it is present only in IDS!

TABLE VII
Octal Table of Two-digit Numbers

IDS equivalent	Octal No.	New figure	Read new name as
9	11	11	Read as one-one, because 11 is not eleven! There is no eleven in OS. Eleven is there again only in IDS.
10	12	12	Read as one-two
11	13	13	Read as one-three
...	...	...	...
38	46	46	Read as four-six
...	...	...	...

the era of the great Greek thinkers like Aristotle. It was used for explaining logic. George Boole later on revived it in the 19th century. This system takes for granted only two digits—the present one and the absent one. The present one is called by various names such as true, yes,

BS just tailor-made in developing the logic circuits.

In comparison, we find that if the IDS tells us as to how many different levels a number can be present in, in a given place, the BS using only two digits indicates just whether a place value is present or not.

high, '1' and positive. Its value is equal to '1' in the IDS. The absent one is the same as our zero and is called by various names indicating the opposites of the former. In the second half of the present century, vacuum-tube gate circuits found the

Here are some features of BS:

1. There are only two digits: 0 and 1

2. Higher-order numbers are written *a la* the IDS

Table III briefly highlights these features.

OS. The octal system (OS), the fourth one, falls short of the IDS by two digits, with counting ending at '7' itself. For '8' of the IDS, we write '10' in the OS. Higher orders are written as in the IDS

HdS. The hexadecimal system (HdS) is the last and giant among all the five counting systems. It fell short of figures for its digital notations. Hence, it borrows for its digits partly from the IDS and partly from the Roman alphabets.

The first ten digits of the HdS are the same as that of IDS, but the letters 'A' through 'F' represent the remaining six. These are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E and F.

(Notice that 'C' in RS = 100, whereas C = 12 in the HdS. 'D' in RS = 500, whereas D = 13 in the HdS. '15' is simply 'F' in the HdS and '16' is '10'.)

This then constitutes the preliminary survey of the different counting systems in use.

The proposition

The IDS and the RS are written uniquely and read uniquely too. So one who hears them as well as the one who sees them, both come to know of the system of counting that the numbers being called or written belong to.

Like, '3' is read as T H R E E in the IDS, whereas 'III' is read as T H I R D in the RS. This feature of distinguishability, however, is limited only to these two systems and does not extend to the other three.

The four-fold aim of this article therefore is to streamline all the five counting systems as follows:

Rule 1 (the audio test). To give a listener understanding of all the five systems, particularly the remaining three, viz, the BS, the OS and the HdS.